1. R7.4: King §5 and Eagle–Galbraith–Ong, and a binary-Edwards check of Corollary 9.2

Date 2026-10-31. Items 1 and 2 of the three small open items of §19.17. Toy instances only (n = 9 … 14). No sub-rho claim and no claim about a deployed curve. Source of every number: `binary_edwards.json` (measured, native Rust `examples/r7_4_binary_edwards.rs`) or the cited page. Full text and the paste-ready note text are in `RESULTS.md`.

1.1. Verdicts

Item 1 (literature). The primary texts of King (PKC 2004) and of Eagle–Galbraith–Ong (EGO) could not be read: Springer redirects to a login and the IACR PDFs return a Cloudflare 403. Read: the ePrint 2009/086 abstract page, the publisher page of EGO (abstract, 13-entry reference list with King as [4]), Eagle's 2008 PhD thesis (a different document; Lemma 4.1.15, $\mathrm{Tr}(x) = \mathrm{Tr}(a)$ on an odd-prime-order subgroup; no King, no halving, no cofactor-4 test), and citation metadata. None of it contradicts or pre-empts a statement of the note, and none of it bears on the kernel-saturation law, symmetrisation or summation polynomials. **King §5 and the EGO cofactor-4 test remain unread.**

Item 2a (algebraic). Proved: Corollary 9.2 holds for every rational subgroup of 2-power order on every ordinary binary curve in every model, so for binary Edwards curves and for Z/4 (Proposition 9.3 in `RESULTS.md`). Tested on 46 toy binary Edwards curves (151 034 points): 11 identity families, 0 exceptions. Binary Edwards has no torsion that Weierstrass lacks: $4 \mid N$ iff $\mathrm{Tr}(d_1) = 1$, as $4 \mid N$ iff $\mathrm{Tr}(a) = 0$ for Weierstrass (1 600 random curves each, 0 exceptions). Its coordinate $w = x + y$ has the orbits of $\langle T, -1\rangle$ as fibres (37 792 fibres), so the natural Edwards frame is T-symmetrised by construction.

Item 2b (pair-table yield). Hypothesis: the yield-ratio interval contains 1. Falsifier as worded: an interval excluding 1 on some family. **The falsifier fires at n ≤ 12** (primary metric: 39 of 40 pooled cells), with a sign that depends on whether the relation matrix is charged c or $c + 1$ rows; I do not confirm the hypothesis at toy size and I do not claim a gain. The naive full sweep reproduces the law $1/\mu$ (n = 13, 14), and the quotient sweep finds the same relations with $1/\mu$ lookups (268 000 of 268 000 checks).

1.2. Method (item 2b)

Complete binary Edwards curves $d_1(x + y) + d_2(x^2 + y^2) = xy + xy(x + y) + x^2y^2$, $\mathrm{Tr}(d_2) = 1$, over $\mathbb{F}_{2^n}$: 40 curves with n = 9 … 12 and 6 with n = 13, 14 (scale check). Cyclic group of order N, r the largest prime factor ($r \geq 100$, $r^2 \nmid N$). Pair-table oracle exact over every target of G; randomness only in 1 000 bases per arm per cell. Arms at equal $|F|$: plain, s2 (saturated under $\langle T\rangle$, the fibres of w), s4 (under $\langle Q_4\rangle$, if $4 \mid N$), each swept over all of F or over one point per coset (quotient). Yield $= \sum \lambda / \sum (UW)$, λ the mean number of relations modulo M per target, W lookups, $U = c$ or $c + 1$ for c distinct signed projected columns; the primary metric is λ/c. Interval: 2 000 bootstrap resamples over bases, stratified by curve.

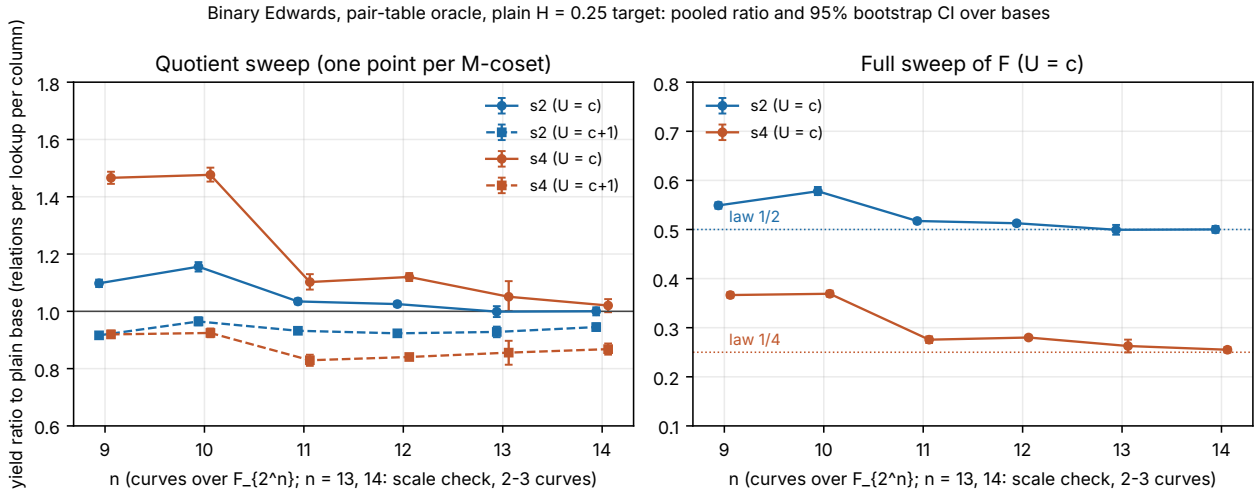


Figure 1: Pooled yield ratio to the plain base and 95% bootstrap interval against n, plain hit fraction target 0.25 (`binary_edwards.json`, `pooled_ratios`; n = 13, 14: 3 curves each, 2 and 3 with $4 \mid N$). Left: quotient sweep, $U = c$ (solid) and $U = c + 1$ (dashed). Right: full sweep with the law $1/\mu$ (dotted).

1.3. Results

Pooled over the 40 curves with n ≤ 12 (ratio, 95% interval).

arm, metric	H 0.05	0.1	0.25	0.5
s2 quot, λ/c	1.119 [1.110, 1.128]	1.084 [1.077, 1.091]	1.061 [1.056, 1.067]	1.048 [1.044, 1.050]
s2 quot, λ/(c+1)	0.934 [0.926, 0.941]	0.942 [0.936, 0.948]	0.940 [0.934, 0.945]	0.963 [0.960, 0.966]
s4 quot, λ/c	1.481 [1.464, 1.500]	1.342 [1.329, 1.354]	1.212 [1.198, 1.228]	1.109 [1.103, 1.116]
s4 quot, λ/(c+1)	0.928 [0.918, 0.940]	0.924 [0.915, 0.933]	0.880 [0.870, 0.891]	0.884 [0.878, 0.889]
s2 full, λ/c (law 1/2)	0.559 [0.555, 0.564]	0.542 [0.538, 0.545]	0.530 [0.528, 0.533]	0.524 [0.522, 0.525]
s4 full, λ/c (law 1/4)	0.370 [0.366, 0.374]	0.336 [0.333, 0.338]	0.303 [0.299, 0.307]	0.277 [0.276, 0.279]

Scale check, n = 13 and 14, primary metric: s2 quot 0.992–1.038, s4 quot 0.994–1.146; the interval contains 1 in 10 of 16 cells; s2 full 0.496–0.519 and s4 full 0.248–0.286, the law value inside the interval in 12 of 16 cells. The hit-fraction metric H/c keeps an excess (1.14 at n = 14, H target 0.05, s2) that decreases with n and that I cannot attribute; larger instances would be needed.

1.4. Limits

Toy $n \leq 14$ with 1–20 projected columns (R5.1 found the law unreached in this regime). The pair-table yield depends on the abstract group and the base, not on the model; no Gröbner/F4 oracle was run on Edwards summation polynomials, and the Galbraith–Gebregiyorgis and Huang–Petit–Shinohara–Takagi setups were not read. The primary metric was fixed after a smoke run; all four are in the JSON. The finite-size reading of the c versus $c + 1$ split is derived, not proved.

1.5. Unread sources

King PKC 2004 (§5, Table 2); the EGO body; Pornin ePrint 2022/1164 and 2022/1325; Galbraith–Gebregiyorgis 2014; Huang–Petit–Shinohara–Takagi 2015. A search-engine summary says EGO refers to King §5 for the cofactor-4 case; that is snippet-level and unverified.